

Single-Arc Phase Compression Does Not Explain the Production Horizons

Spectral-Measure Krylov Identities, Arc Polynomial Bounds, and a Frozen-Schur Negative Audit

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July 2026

Abstract

RH-75 reduced the uniform directional-Hardy problem to a log-square horizon and square-root block-contraction law. One possible mechanism was that the physical source becomes concentrated on a shrinking spectral phase arc. This paper derives exact tests for that mechanism and finds that it fails for the production Schur surrogates.

For a unitary U and source x with spectral measure μ_x ,

$$\text{dist}(U^M x, \mathcal{K}_d(U, x))^2 = \inf_{\deg p < d} \int_{\mathbb{T}} |z^M - p(z)|^2 d\mu_x(z).$$

If the support lies in an arc of width w , truncating the binomial expansion around its center gives the explicit degree- $(d-1)$ upper

$$\sum_{j=d}^M \binom{M}{j} \{2 \sin(w/4)\}^j.$$

Conversely, a moment-coherence bound gives

$$\text{dist}^2 \geq 1 - \frac{d\eta^2}{1 - (d-1)\eta}.$$

These formulas quantify how narrow an arc, or how coherent a moment sequence, must be to reduce depth below $M+1$.

We form source-weighted phase measures from the frozen production Schur forms. The displayed binary64 phases and weights are then embedded as exact dyadic centers, and 192-bit Arb moment Gramians certify the projection residual at depth M . The 99% arc widths grow toward the full circle. At $\sigma = 0.01$ they are 5.9034 and 5.8465 radians, while the certified depth-32 residuals are above 0.7709 and 0.6114. Both channels require the full monomial depth 33 even for 10% error. Across all scales, nine of ten channels require full depth for 10%, and all ten for 1%.

Thus single-arc phase compression is a closed negative branch. This does not invalidate the RH-75 scaling law: broad phase support may still have low weighted effective rank, multi-arc packet structure, or rapid radial decay. RH-77 pursues that fallback. The Schur construction remains a frozen diagnostic, so no continuum phase-measure theorem or Stage A1 closure is claimed.

Keywords: Krylov subspace; spectral measure; phase compression; moment Gramian; negative result; interval arithmetic.

MSC 2020: 47A10; 47B15; 65F10; 65G20.

1 Introduction

The exact Fourier-ring construction of RH-68 showed that broad coherent phase support can force full Krylov depth even under perfect metric conditioning [3]. RH-75 nevertheless left open a physical escape: perhaps the production source occupies a shrinking phase arc, unlike the adversarial full ring [4].

There are two questions:

1. What quantitative arc width is sufficient for depth reduction?
2. Do the source-weighted production phases exhibit that width?

The first is answered exactly for normal phase packets. The second is tested on the same frozen Schur data used by the phase-aware Hardy papers. The answer is negative: the weighted phase support becomes broader, not narrower.

This is intentionally a branch-level result. A nonnormal system can gain from radial decay, multiple packets, source/observation cancellation, or singular value decay even when one global phase arc is broad. The audit only rules out the simplest single-arc explanation.

2 Spectral-measure Krylov identity

Let U be unitary on a finite-dimensional Hilbert space, let x be a unit vector, and let μ_x be its spectral measure on $\mathbb{T}$. Write

$$\mathcal{K}_d(U, x) = \text{span}\{x, Ux, \dots, U^{d-1}x\}.$$

Theorem 2.1 (Phase-measure projection identity). *For $M \geq d \geq 1$,*

$$\text{dist}(U^M x, \mathcal{K}_d(U, x))^2 = \inf_{\deg p < d} \int_{\mathbb{T}} |z^M - p(z)|^2 d\mu_x(z). \quad (1)$$

If $G_{ab} = \hat{\mu}_x(b - a)$ and $c_a = \hat{\mu}_x(M - a)$ for $0 \leq a, b < d$, then, whenever G is invertible,

$$\text{dist}^2 = 1 - c^* G^{-1} c. \quad (2)$$

Proof. The spectral theorem identifies $U^j x$ with z^j in $L^2(\mu_x)$. Orthogonal projection onto the first d monomials gives (1); the normal equations give (2). $\square$

This identity is standard Krylov polynomial approximation in spectral-measure form [2]. It also provides a small interval certificate: one needs only the moment Gramian, not the full operator.

3 What a narrow arc would buy

Suppose the support lies in an arc centered at $z_0 \in \mathbb{T}$ with angular width w . Then $|z - z_0| \leq \delta_w := 2 \sin(w/4)$.

Proposition 3.1 (Arc binomial Krylov upper). *For $1 \leq d \leq M + 1$,*

$$\text{dist}(U^M x, \mathcal{K}_d(U, x)) \leq \sum_{j=d}^M \binom{M}{j} \delta_w^j. \quad (3)$$

Proof. Expand $z^M = z_0^M (1 + (z - z_0)/z_0)^M$ and retain orders below d . The retained part is a polynomial of degree below d in z ; bound the remainder uniformly on the arc and use (1). $\square$

Stirling's bound shows the familiar scale $d \gtrsim eM\delta_w + \log(1/\varepsilon)$. With $M = O(k^2)$, a depth $O(k)$ would therefore require $w = O(k^{-1})$. An arc of fixed positive width does not explain logarithmic compression.

4 Moment-coherence obstruction

Let the monomials $1, z, \dots, z^{d-1}$ be normalized in $L^2(\mu)$ and assume all off-diagonal basis moments and all correlations with z^M have modulus at most η .

Proposition 4.1 (Moment-coherence lower). *If $(d-1)\eta < 1$, then*

$$\text{dist}(z^M, \text{span}\{1, \dots, z^{d-1}\})^2 \geq 1 - \frac{d\eta^2}{1 - (d-1)\eta}. \quad (4)$$

Proof. Gershgorin gives $\lambda_{\min}(G) \geq 1 - (d-1)\eta$, while $\|c\|^2 \leq d\eta^2$. Apply (2). $\square$

The uniform Fourier ring has $\eta = 0$ and residual one until the target monomial itself enters the basis. The proposition makes that obstruction robust.

5 Frozen production phase audit

For each RH-75 scale and each left/right system, we compute a unitary complex Schur form $A = QTQ^*$ and source coordinates $Z = Q^*X$. The diagonal phases $\theta_j = \arg T_{jj}$ receive weights

$$w_j = \|Z_{j,:}\|_2^2.$$

The minimal circular arc containing a prescribed fraction of $\sum_j w_j$ is then found by an exact sorted sliding window on the displayed dyadic centers.

For the moment calculation, the binary64 θ_j, w_j are embedded as exact dyadic numbers. Arb evaluates $e^{i\theta_j}$, assembles the depth- M moment Gramian, solves its normal equations, and encloses $1 - c^*G^{-1}c$ at 192 bits [1]. This certifies the *frozen-Schur normal surrogate*; it does not validate the Schur decomposition against the analytic continuum operator.

Table 1: Source-weighted 99% arc widths and certified residual at depth M .

σ	left		right		M
	width	residual	width	residual	
0.16	3.6723	0.5161	4.6779	0.4499	4
0.08	4.9693	0.3008	5.0513	0.01035	9
0.04	5.3345	0.4774	5.2041	0.1764	16
0.02	5.5462	0.5278	5.8065	0.5989	25
0.01	5.9034	0.7709	5.8465	0.6114	32

At the finest scale, the 99% arcs occupy more than 93% of the full circle. The exact moment-surrogate residual remains far above 10% even after all degrees $0, \dots, M-1$ are used. A direct diagnostic search gives required 10% depths

$$(5, 5), (10, 8), (17, 17), (26, 26), (33, 33),$$

for left/right levels respectively; full depth is $M+1$. At 1%, every entry equals full depth.

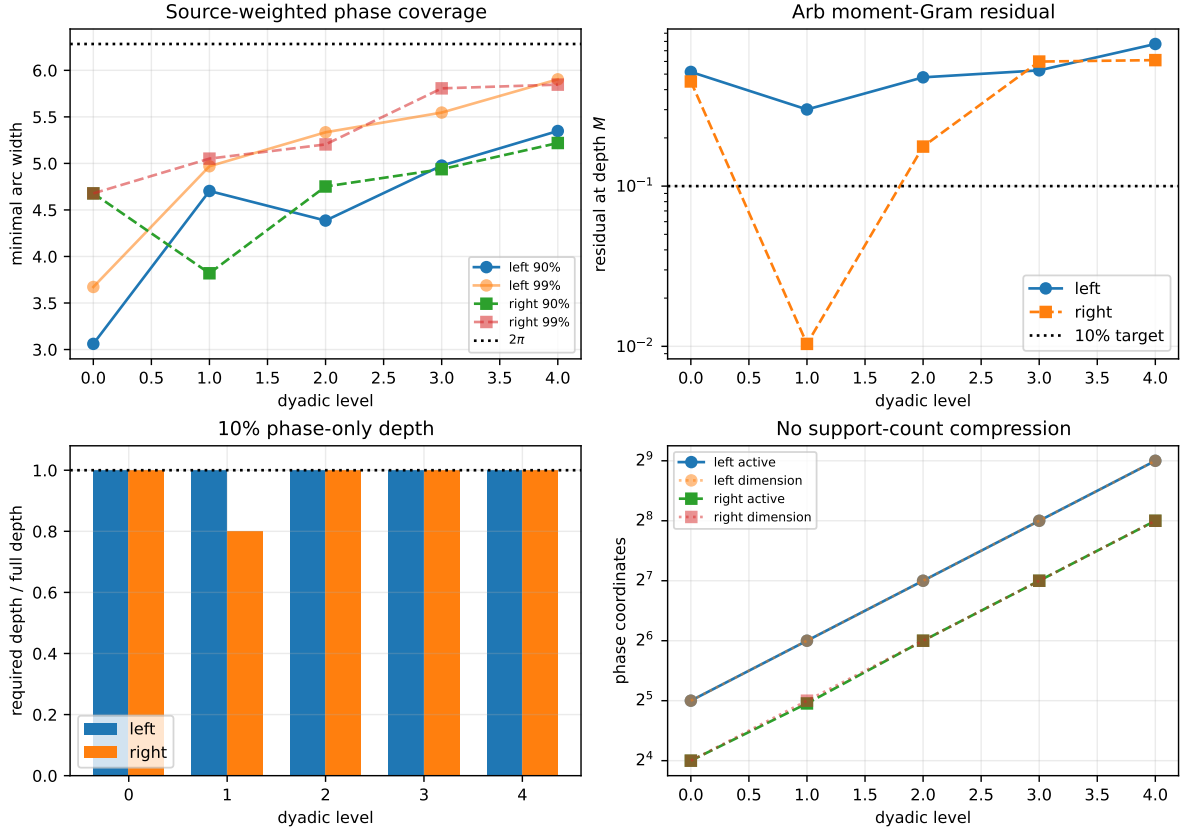


Figure 1: Weighted arc widths, interval moment residuals, required phase-only depth, and active phase-coordinate counts.

6 Route verdict

The single-arc hypothesis predicts shrinking widths of order $1/k$ if it is to turn an $O(k^2)$ horizon into substantially smaller polynomial depth. The frozen production surrogates show the opposite trend. Single-arc phase compression is therefore marked as a failed branch.

What survives is not a cosmetic variant of the same idea. Broad support can still be harmless when:

1. source weights have low effective rank or fast entropy-tail decay;
2. phases split into several packets with small total weighted complexity;
3. radial powers suppress most packets before phase approximation matters;
4. nonnormal cross-packet cancellation is retained rather than replaced by a normal surrogate.

RH-77 tests weighted effective rank and multi-packet decay.

No validated continuum phase measure, uniform phase-compression theorem, Stage A1 closure, unconditional Stage A4, Hilbert–Polya operator, $T \log T$ law, prime-power trace formula, zeta-zero identity, or proof of the Riemann Hypothesis is obtained.

7 Conclusion

The maze has a clearly marked wall: one shrinking phase arc is not the source of the observed production horizons. The negative result is useful because it prevents the RH-75 all-level target

from being tied to a false mechanism. The remaining path is weighted rather than geometric: effective rank, multi-arc packets, and radial damping.

References

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